

Course Name: General Physics B (II) Department: PHYSICSExam Duration: 2 hours Exam Paper Setter: Physics teaching team

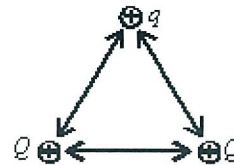
Question No.	1	2	3	4	5	6	7	8	9	10
Score	30	6	10	10	12	10	10	12		

This exam paper contains 8 questions and the score is 100 in total. (Hand in your exam paper, answer sheet, and draft paper to the proctor when the exam ends.)

Q1. Multiple Choice Questions ($3 \times 10 = 30$ scores), each question has one correct answer.

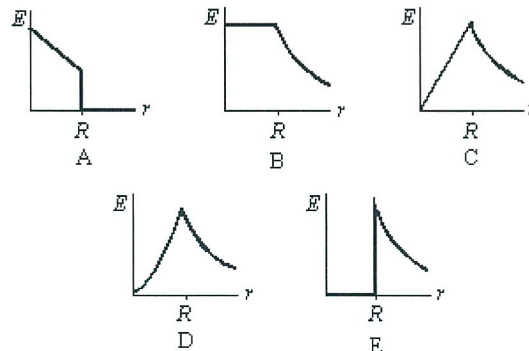
1. Two particles, each with charge Q , and a third particle, with a charge q , are placed at the vertices of an equilateral triangle as shown. The total force on the particle with charge q is:

- A) Parallel to the left side of the triangle
- B) Parallel to the right side of the triangle
- C) Parallel to the bottom side of the triangle
- D) Perpendicular to the bottom side of the triangle
- E) Perpendicular to the left side of the triangle



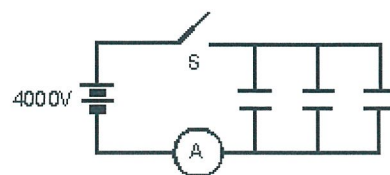
2. Which of the following graphs represents the magnitude of the electric field as a function of the distance from the center of a solid **charged conducting** sphere of radius R ?

- A) A
- B) B
- C) C
- D) D
- E) E



3. Each of the three $25\text{-}\mu\text{F}$ capacitors shown is initially uncharged. How many coulombs of charge pass through the ammeter A after the switch S is closed?

- A) 0.033 C
- B) 0.10 C
- C) 0.30 C
- D) 10 C
- E) none of these



4. The diagram shows three equally spaced wires that are perpendicular to the page. The currents are all equal, two being out of the page and one being into the page. Rank the wires according to the magnitudes of the magnetic forces on them, from least to greatest.

- A) 1, 2, 3
 B) 2, then 1 and 3 tie
 C) 2 and 3 tie, then 1
 D) 1 and 3 tie, then 2
 E) 3, 2, 1



5. A 10-turn ideal solenoid has an inductance of 4.0 mH. To generate an emf of 2.0 V the current should change at a rate of:

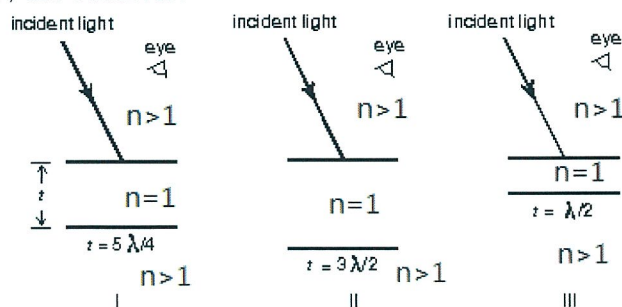
- A) 500 A/s
 B) 250/s
 C) 50 A/s
 D) 0.5/s
 E) 0 A/s

6. Light with an intensity of 1 kW/m^2 falls normally on a surface and is completely absorbed. The radiation pressure is:

- A) 1 kPa
 B) $3 \times 10^{11} \text{ Pa}$
 C) $1.7 \times 10^{-6} \text{ Pa}$
 D) $3.3 \times 10^{-6} \text{ Pa}$
 E) $6.7 \times 10^{-6} \text{ Pa}$

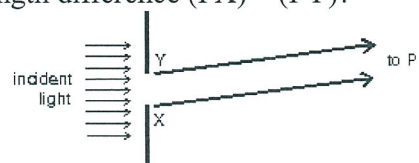
7. Three experiments involving a thin film are shown. If t denotes the film thickness and λ denotes the wavelength of the light in the film, which experiments will produce constructive interference as seen by the observer?

- A) I and III only
 B) II and III only
 C) I only
 D) II only
 E) III only



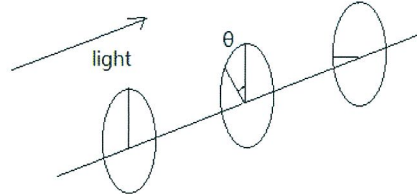
8. The diagram shows a single slit with the direction to a point P on a distant screen (not shown). At P, the pattern has its second minimum (from its central maximum). If X and Y are the edges of the slit, what is the path length difference (PX) – (PY)?

- A) $\lambda/2$
 B) λ
 C) $3\lambda/2$
 D) 2λ
 E) $\lambda/2$



9. Three polarizing sheets are placed in a stack with the polarizing directions of the first and third perpendicular to each other. What angle should the polarizing direction of the middle sheet make with the polarizing direction of the first sheet to obtain maximum transmitted intensity when unpolarized light is incident on the stack?

- A) 0°
 B) 30°
 C) 45°
 D) 60°
 E) 90°



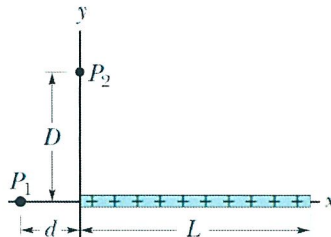
10. A meter stick moves sideways (that is, in a direction perpendicular to its length) at $0.95c$. According to measurements taken in the laboratory, its length is:

- A) 0 m
 B) 0.098 m
 C) 0.31 m
 D) 1.0 m
 E) 3.2 m

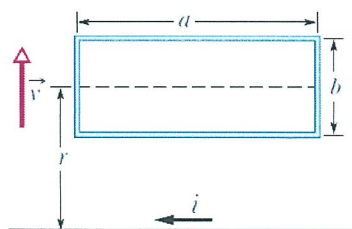
Essays questions

Q2. (6 scores) The wavelength of the yellow spectral emission line of sodium is 600 nm. At what kinetic energy would an electron have that wavelength as its de Broglie wavelength?

Q3. (10 scores) The thin plastic rod of length $L = 12.0$ cm has a nonuniform linear charge density $\lambda = cx$, where $c = 50$ pC/m². (a) With $V = 0$ at infinity, find the electric potential at point P_2 on the y axis at $y = D = 4.0$ cm. (b) Find the electric field component E_y at P_2 .



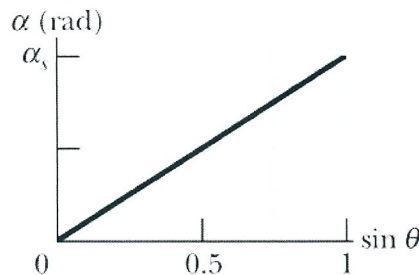
Q4. (10 scores) In the figure, a rectangular loop of wire with length $a = 2.2$ cm, width $b = 0.80$ cm, and resistance $R = 0.40$ m Ω is placed near an infinitely long wire carrying current $i = 6.9$ A. The loop is then moved away from the wire at constant speed $v = 3.2$ mm/s. When the center of the loop is at distance $r = 1.5b$, what are (a) the magnitude of the magnetic flux through the loop and (b) the current induced in the loop?



Q5. (12 scores) An alternating emf source with a variable frequency f_d is connected in series with a 50.0 resistor and a 28.0 mF capacitor. The emf amplitude is 12.0 V. (a) At what driving frequency f_d do the two phasors have the same length? At that driving frequency, what are (b) the phase angle in degrees, (c) the angular speed at which the phasors rotate, and (d) the current amplitude?

Q6. (10 scores) In the Young double-slit experiment, the viewing screen is at distance $D = 4.00$ m, point P lies at distance $y = 22$ cm from the center of the pattern, the slit separation d is $4.50 \mu\text{m}$, and the wavelength λ is 650 nm. (a) Determine where point P is in the interference pattern by giving the maximum or minimum on which it lies, or the maximum and minimum between which it lies. (b) What is the ratio of the intensity I_P at point P to the intensity I_{cen} at the center of the pattern?

Q7. (10 scores) Figure gives α versus the sine of the angle θ in a single-slit diffraction experiment using light of wavelength 610 nm. The vertical axis scale is set by $\alpha_s = 24$ rad. What are (a) the slit width, (b) the total number of diffraction minima in the pattern (count them on both sides of the center of the diffraction pattern), (c) the least angle for a minimum, and (d) the greatest angle for a minimum?



Q8. (12 scores) A space traveler takes off from Earth and moves at speed $0.99c$ toward the star Vega, which is 26.00 ly distant. How much time will have passed by Earth clocks (a) when the traveler reaches Vega and (b) when Earth observers receive word from the traveler that she has arrived? (c) How much older will Earth observers calculate the traveler to be (measured from her frame) when she reaches Vega than she was when she started the trip?



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Q1. Multiple Choice Questions ($3 \times 10 = 30$ scores)

1-5 DECBA

6-10 DADCD

Q2

Solution :

If K is given in electron volts, then

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}} = \frac{6.626 \times 10^{-34} \text{ J}\cdot\text{s}}{\sqrt{2(9.109 \times 10^{-31} \text{ kg})(1.602 \times 10^{-19} \text{ J/eV})K}} = \frac{1.226 \times 10^{-9} \text{ m}\cdot\text{eV}^{1/2}}{\sqrt{K}}$$

$$= \frac{1.226 \text{ nm}\cdot\text{eV}^{1/2}}{\sqrt{K}},$$

...4'

where K is the kinetic energy. Thus,

$$K = \left(\frac{1.226 \times 10^{-9} \text{ m}}{\lambda} \right)^2 \text{ eV} = \left(\frac{1.226 \text{ nm}}{600} \right)^2 \text{ eV} = 4.18 \times 10^{-6} \text{ eV} \quad \dots 2'$$

Q3.

Solution:

(a) Consider a charge element of the rod from x to $x + dx$. Its contribution to the potential at point P_2 is

$$dV = \frac{1}{4\pi\epsilon_0} \frac{\lambda(x)dx}{\sqrt{x^2 + y^2}} = \frac{1}{4\pi\epsilon_0} \frac{cx dx}{\sqrt{x^2 + y^2}}$$

$$V = \int dV_p = \frac{c}{4\pi\epsilon_0} \int_0^L \frac{xdx}{\sqrt{x^2 + y^2}} = \frac{c}{4\pi\epsilon_0} (\sqrt{L^2 + y^2} - y)$$

$$= 3.88 \times 10^{-2} \text{ V} \quad \dots 6'$$

(b) The y component of the field is

$$E_y = -\frac{\partial V_p}{\partial Y} = -\frac{c}{4\pi\epsilon_0} \frac{d}{dy} (\sqrt{L^2 + y^2} - y) = \frac{c}{4\pi\epsilon_0} \frac{d}{dy} \left(1 - \frac{y}{\sqrt{L^2 + y^2}}\right) \quad \dots 4'$$

$$= 0.307 N/C$$

Q4

Solution :

(a) We assume the flux is entirely due to the field generated by the long straight wire

$$|\Phi_B| = \int_{r-b/2}^{r+b/2} \left(\frac{\mu_0 i}{2\pi r} \right) (a dr) = \frac{\mu_0 i a}{2\pi} \ln \left(\frac{r+b/2}{r-b/2} \right).$$

When $r = 1.5b$, we have

$$|\Phi_B| = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(6.9 \text{ A})(0.022 \text{ m})}{2\pi} \ln(2.0) = 2.1 \times 10^{-8} \text{ Wb}. \quad \dots 5'$$

(b) Implementing Faraday's law involves taking a derivative of the flux in part (a), and recognizing that $dr/dt = v$. The magnitude of the induced emf divided by the loop resistance then gives the induced current:

$$i_{\text{loop}} = \left| \frac{\mathcal{E}}{R} \right| = \left| -\frac{\mu_0 i a}{2\pi R} \frac{d}{dt} \ln \left(\frac{r+b/2}{r-b/2} \right) \right| = \frac{\mu_0 i a b v}{2\pi R [r^2 - (b/2)^2]} = \frac{\mu_0 i a b v}{2\pi R (2b^2)}$$

$$= \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(6.9 \text{ A})(0.022 \text{ m})(0.0080 \text{ m})(3.2 \times 10^{-3} \text{ m/s})}{2\pi (4.0 \times 10^{-4} \Omega) [2(0.0080 \text{ m})^2]} \quad \dots 5'$$

$$= 1.52 \times 10^{-5} \text{ A}.$$

Q5.

Solution:

(a) We have $1/R = 1/X_C$, $R = \frac{1}{\omega_d C}$

which yields $f = \frac{\omega_d}{2\pi} = \frac{1}{2\pi RC} = \frac{1}{2\pi(50.0 \Omega)(2.80 \times 10^{-5} \text{ F})} = 114 \text{ Hz}. \quad \dots 3'$

(b) $\phi = \tan^{-1}(-V_C/V_R) = -45^\circ \quad \dots 3'$

(c) $\omega_d = 1/RC = 7.14 \times 10^2 \text{ rad/s}. \quad \dots 3'$

(d) $I = 12 \text{ V} \frac{1}{\sqrt{R^2 + X_C^2}} = 170 \text{ mA} \quad \dots 3'$

Q6.

Solution:

(a) $\theta = \tan^{-1}(y/D) = \tan^{-1}(0.22/4) = 3.15^\circ \quad \dots 4'$

Thus, the phase difference at point P is $\phi = 2\pi d \sin \theta / \lambda = (0.3808)(2\pi) = 2.393 \text{ rad}$, which

means it is between the central maximum (0λ difference) and the first minimum ($\frac{1}{2}\lambda$ difference). Note that the above computation could have been simplified somewhat by avoiding the explicit use of the tangent and sine functions and making use of the small-angle approximation ($\tan\theta \approx \sin\theta$).

(b) $\phi = (0.3808)(2\pi) = 2.393 \text{ rad}$

At point p, $I = 4I_0 \cos^2(\phi/2) = 0.535I_0$. At the center, $I_{\text{center}} = 4I_0 \cos^2(0) = 4I_0$

Thus, $I/I_{\text{center}} = 0.785/4 = 0.134$6'

Q7.

Solution

(a) The slope of the plotted line is 12, and this slope should correspond to

$$\frac{\pi a}{\lambda} = 24 \Rightarrow a = \frac{24\lambda}{\pi} = \frac{24(600 \text{ nm})}{\pi} = 4583 \text{ nm} \approx 4.583 \mu\text{m} \quad \text{...3'}$$

(b) Consider “continuously variable” m (of course, m should be an integer for diffraction minima, but for the moment we will solve for it as if it could be any real number):

$$m_{\text{max}} = \frac{a}{\lambda} (\sin\theta)_{\text{max}} = \frac{24\lambda}{\pi\lambda} = \frac{24}{\pi} \approx 7.6 \quad \text{...3'}$$

which suggests that, on each side of the central maximum ($\theta_{\text{centr}} = 0$), there are **7 minima**; considering both sides then implies there are **14 minima** in the pattern.

(c) Setting $m = 1$ and solving for θ yields 7.5°2'

(d) Setting $m = 7$ and solving for θ yields 66.4°2'

Q8.

Solution:

(a) The speed of the traveler is $v = 0.990c$, which may be equivalently expressed as 0.990ly/y . Let d be the distance traveled. Then, the time for the trip as measured in the frame of Earth is

$$\Delta t = d/v = (26 \text{ ly})/(0.99 \text{ ly/y}) = 26.26 \text{ y.}$$

(b) The signal, presumed to be a radio wave, travels with speed c and so takes 26.0 y to reach Earth. The total time elapsed, in the frame of Earth, is

$$26.26 \text{ y} + 26.0 \text{ y} = 52.26 \text{ y}$$

(c) The proper time interval is measured by a clock in the spaceship, so $\Delta t_0 = \Delta t/\gamma$. Now

$$\gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{1}{\sqrt{1-(0.99)^2}} = 7.09$$

Thus, $\Delta t_0 = (26.26 \text{ y})/(7.09) = 3.704 \text{ y.}$

